Credit Score Classification using XGBoost

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Abstract. Credit scoring underpins financial decision-making but remains challenged by the trade-off between accuracy & transparency. Traditional models are interpretable yet limited in complexity, while advanced machine learning models offer higher accuracy but reduced explainability. This study develops an explainable credit scoring model combining XGBoost with SHAP analysis. The system employs leak-free preprocessing, hyperparameter tuning, & visual evaluation. Results show strong predictive accuracy & interpretability, supporting ethical, fair, & transparent AI-driven financial decision-making [2, 3].

1 Introduction

With AI, credit risk assessment has been revolutionised to make data-driven predictions of borrower reliability. Traditional statistical models provide interpretability but fail in capturing non-linear financial patterns found in modern credit datasets. On the other hand, machine learning methods like XGBoost, Logistic Regression, KNN, & Neural Networks give better predictive performance but more often act as a "black box" system that raises concerns regarding trust, fairness & regulatory compliance. This study will fill this gap by assessing four algorithms & developing an explainable framework using XGBoost–SHAP. The hypothesis is that this combination can achieve high multi-class accuracy while retaining the transparency required in financial decision-making. [1, 3, 4]

2 Background

Credit scoring is a supervised classification problem where models learn from labeled historical outcomes to predict borrower creditworthiness. Modern machine learning methods have proven to learn non-linear interactions more effectively & among them, gradient boosting models consistently outperform classical techniques. Recent studies report XGBoost achieving AUC values above 0.97 in credit risk prediction. [2]

Yet, these models create an interpretability gap, making fairness, accountability & compliance with regulations like GDPR Article 22 challenging. SHAP provides model-agnostic explanations, enabling transparent reasoning behind predictions. Thus, the integration of XGBoost & SHAP directly meets the existing performance-explainability gap in financial AI. [1, 3]

3 Experiment

The dataset had demographic & financial attributes, with a categorical target variable for Credit Score: Low, Average & High. The data were complete, with no needs for imputation.

Categorical variables were encoded using OneHotEncoder within a ColumnTransformer to avoid imposing false ordinal relationships & to prevent leakage during training. This example does not scale the numerical features since XGBoost has a scale-invariant tree structure. LabelEncoder changed the target categories to numeric classes. This preparation pipeline reflects realistic credit scoring workflows; the features included are common in banking, such as income, house ownership & employment stability. [1, 2]

Implementation

XGBoost was selected for its ability to capture non-linear interactions, handle mixed data types, & incorporate internal regularisation (features that outperform classical models). [1, 2]

A leak-free Pipeline combined the preprocessing with model training to ensure that all transformations occurred consistently within each fold of cross-validation. The classifier used was XGBClassifier with objective="multi:softprob", producing class-probability outputs suitable for multi-class evaluation.

Hyperparameter tuning was performed with RandomizedSearchCV & StratifiedKFold (5-fold) to preserve class balance & reduce variance due to the small dataset size. [1, 2]

Parameter	**Configuration**
$n\_estimators$	89
max_depth	5
$learning\_rate$	0.098
$subsample$	0.824
$colsample\_bytree$	0.774

Table 1. The final optimised configuration

This architecture yielded a well-generalised model that could capture non-linear relationships without excessive depth or instability.

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Table 2. Hyperparameter Roles in the XGBoost Model

Parameter	Role in Model & Contextual Meaning
n_estimators	Number of decision trees (more trees improve accuracy but risk overfitting if excessive).
max_depth	Controls how deeply each tree can grow (shallow trees generalise better; deeper trees capture complex relations).
learning_rate	Determines how much each new tree influences the model (a smaller rate encourages gradual learning & stability).
subsample	Fraction of features used per tree (ensures diverse trees & prevents bias from dominant features).
colsample_bytree	Fraction of features used per tree (ensures diverse trees & prevents bias from dominant features).

Results

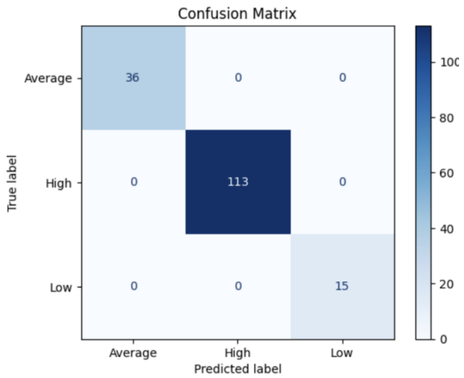

Figure 1. Confusion Matrix – Comparison of Predicted vs True Labels

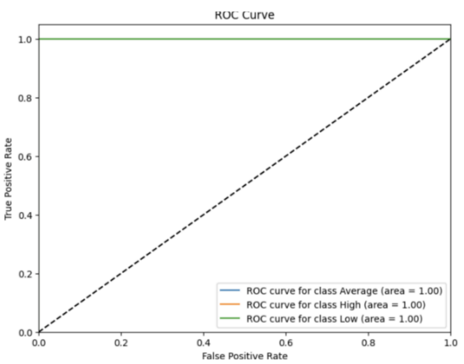

Figure 2. ROC Curves – One-vs-Rest AUC Analysis

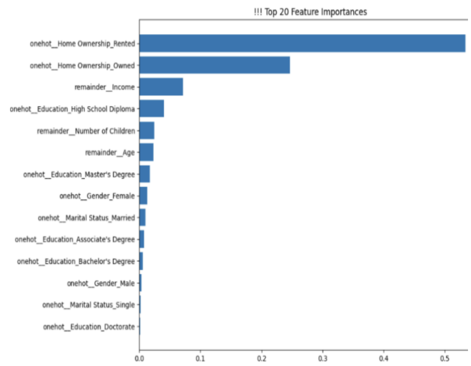

Figure 3. Top 20 Feature Importances – Model-Derived Insights

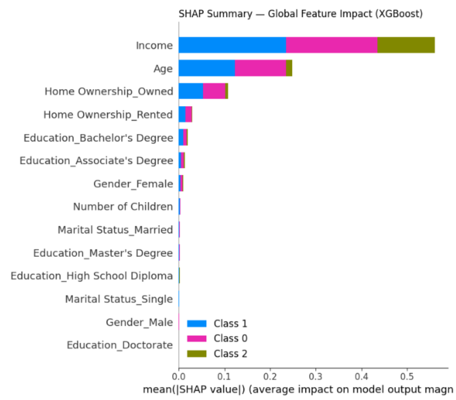

Figure 4. SHAP Summary – Global Feature Impact (Bar Version)

Metric / Visualisation	What the Metric Means	What the Figure Shows for This Model	Why It Is Important (Financial Context)
Confusion Matrix	Shows correct vs incorrect classifications for each class. Highlights directional errors.	Perfect diagonal: 36 Average, 113 High, 15 Low correctly classified. No misclassification.	Avoids catastrophic errors. Essential for safe lending. [1, 5]
ROC Curve (AUC)	Measures discriminative ability across threshold values. AUC=1.00 is perfect.	All classes achieved AUC=1.00, indicating flawless ranking of risk levels.	Helps lenders rank borrowers with maximum reliability; reduces default risk & improves interest rate assignment. [2, 5]
Feature Importance	Shows features with highest contribution based on model's internal structure.	Home ownership & income dominate; aligns with financial theory.	Demonstrates model logic follows real-world economic reasoning. Essential for regulator trust. [1, 5]
SHAP Summary (Global Impact)	Explains how each feature affects predictions; ensures transparency.	Income, Age & Home Ownership are the strongest drivers of credit score predictions	Supports GDPR Article 22 by making decision logic human-interpretable; reduces bias & black-box risk. [3, 4]

Table 3. Interpretation of Evaluation Metrics & Visualisations

4 Discussion

Comparison With Published Results

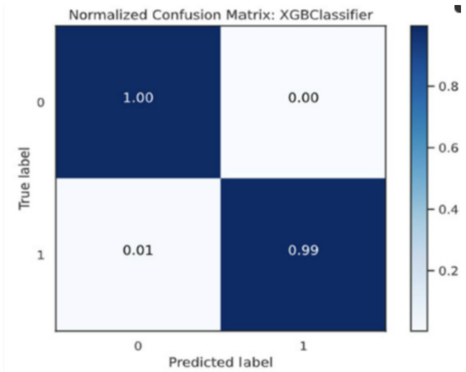

Figure 5. Published CM for credit risk algorithm using XGBoost[2]

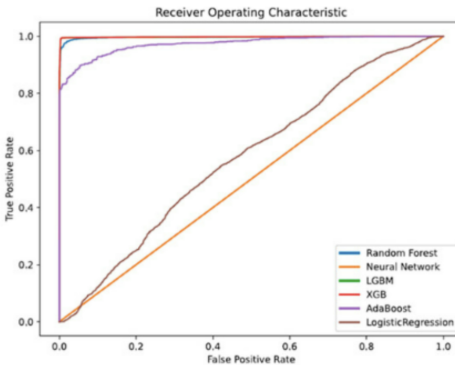

Figure 6. Published ROC for credit risk algorithm[2]

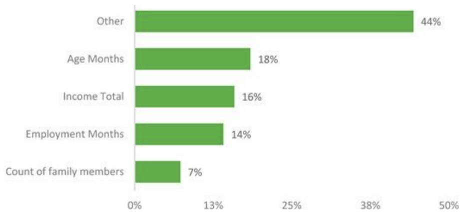

Figure 7. Published FI for credit risk algorithm using XGBoost[2]

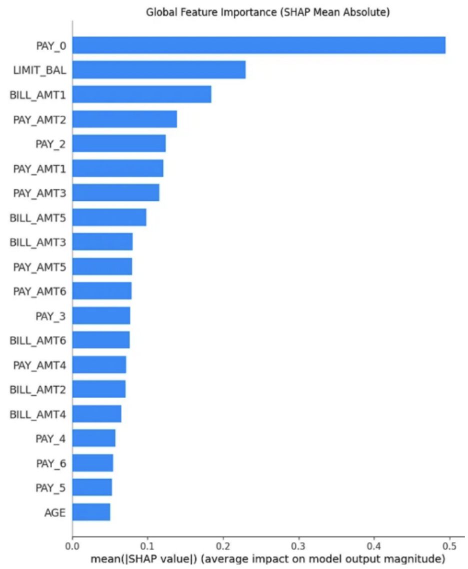

Figure 8. Published SHAP for credit risk algorithm using XGBoost[1]

Visual Type	**Difference**	**Strength**	**Limitation**
CM	Accuracy: mine = 1.00 vs. published = 0.99.	Strong classification & model learning from features.	May reflect overfitting/ dataset bias; external validation required.
ROC	AUC: mine = all classes 1.00 vs. published = 0.95–0.97.	Excellent discriminative ability & reliability across thresholds.	Perfect AUC may result from small/ homogenous data.
FI	Similar ranking to published results..	Confirms interpretability/ consistency of financial theory.	Tree-based importance may overvalue frequent categorical splits.
SHAP	Consistent with published SHAP (strongest influence= income & home ownership).	Provides transparent, ethical explainability in line with GDPR Article 22.	Shows only global impact, local interpretability not visualised.

Comparison of Model Evaluation Metrics with Published Results

LSEPI

[3, 4, 5]

Dimension	Issue / Risk	Impact	Mitigation in This Study
Legal	Automated decision-making engages GDPR Art. 22 & EU AI Act.	Requires transparency & human oversight.	SHAP provides explainability; interpretable workflow supports auditability.
Social / Ethical	Algorithmic bias, exclusion risks, unequal credit allocation.	Risk of reinforcing historic inequalities.	Model aligns with transparent, theory-driven features; avoids opaque black-box decisions.
Professional	Need for reliable ML practice, documentation, & fairness testing.	Prevents automation bias & unsafe deployment.	Pipeline design ensures reproducibility, leakage prevention & accountable modelling.
Practical	Real-world datasets are complex & imbalanced.	Impacts generalisability.	Model design handles categorical diversity; future work includes fairness & stress-testing.
Individual	Applicants require understandable explanations of their credit score.	Supports trust & financial inclusion.	SHAP explanations provide human-readable reasoning for each decision.

Table 4. LSEPI Evaluation Summary

5 Conclusion

This paper proposed an explainable XGBoost credit scoring model that reached a strong predictive accuracy with a transparent explanation using the SHAP analysis. The results displayed clear classification boundaries & financially meaningful feature contributions. Larger, real-world datasets should be involved in future work, together with fairness evaluation & extended validation strategies to ensure that AI-driven credit scoring is reliable, unbiased, & ethically aligned. [3, 4, 5]

REFERENCES

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